



Apache DevLake

A Decision-Maker's Guide | Eldrick Wega · Microsoft

What this guide covers

Part 1 — Making the decision. When DevLake fits and when it doesn't

Part 2 — Understanding DevLake. What it is, what you get, how the architecture works, core concepts, costs

Part 3 — Adopting DevLake. A phased path from laptop to production, plus the dashboards you'll actually see

Part 4 — Reference Architectures. Different ways to splice/dice DevLake data for your organization

For engineering leaders, DevEx/platform teams, and architects evaluating Apache DevLake.



Part 1

Is DevLake right for you?

When DevLake fits — and when it doesn't

Fits well when...

- You've rolled out AI tools (e.g. GitHub Copilot) and leadership wants to see measurable impact on delivery
- Your toolchain is a mix of tools: GitHub, Azure DevOps, Jira, Jenkins, GitLab, or Bitbucket
- You have a DevEx, platform, or engineering productivity team to operate it
- You want to own the data and extend with SQL
- You're comfortable running self-hosted Grafana
- Early stages of building a DevEx program

Less suitable when...

- You have no CI/CD or incident signals (DORA depends on both)
- You need a fully managed SaaS with no operational overhead
- Your stack is primarily niche tools with no DevLake connector
- You want out-of-box insights with zero configuration
- You have an existing data lake or existing custom data pipelines created from your tools

DevLake in context

Capability	Apache DevLake	GitHub built-ins	Commercial DevEx <i>(LinearB, Jellyfish, Swarmia, DX)</i>
Copilot usage data	Yes (normalised)	Yes (native)	Via integrations
DORA metrics	Built-in	No	Yes
Cross-tool correlation	Yes (SQL)	No	Yes (proprietary)
SQL-level extensibility	Full	Limited	Limited
Cost model	OSS + infra (~\$30–50/mo)	Included	\$ per seat

Not a feature scorecard — a shape-of-landscape view. Choose based on ownership model, data access, and your team’s extensibility needs.



Part 2

Understanding DevLake

What Apache DevLake is

Open source. An Apache Software Foundation (Incubating) project — not a vendor product.

A dev data warehouse, not a dashboard. It ingests raw data from your DevOps tools into a normalised schema, then visualises it through Grafana.¹

20+ connectors. GitHub, Azure DevOps, Jira, Jenkins, GitLab, Bitbucket, SonarQube, PagerDuty, CircleCI, Copilot, and more.

Fully SQL-queryable. Every panel is backed by a SQL query you can inspect, modify, or extend.

Dashboards out of the Box

DORA dashboards

Deployment Frequency · Lead Time for Changes · Change Failure Rate · Mean Time to Restore.

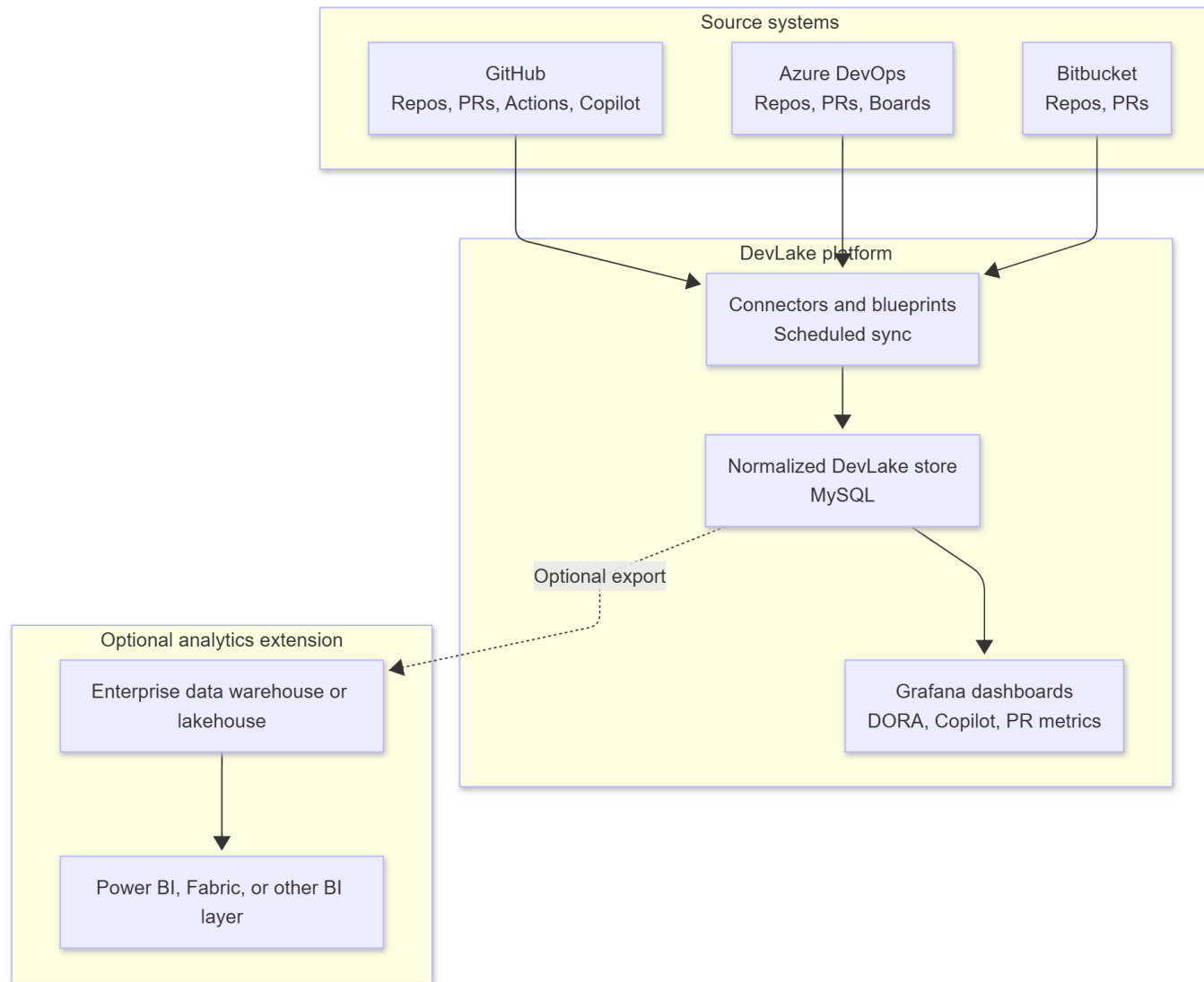
Built-in Elite/High/Medium/Low benchmarking against DORA research.

Copilot Adoption

30+ panels — DAU/WAU/MAU, acceptance rates, language/editor/model breakdowns, seat utilisation, feature mix (completions vs chat vs PR summaries).

Copilot Impact (Microsoft-built)

Correlates adoption intensity (<25%, 25–50%, 50–75%, >75%) with DORA: PR cycle time, deployment frequency, CFR, MTTR, review time.



How it works

DevLake ingests data from your DevOps tools through connectors, normalises it into a MySQL data lake, and exposes it via Grafana dashboards.

Optional: export to an enterprise data warehouse to blend engineering data with business/financial signals in Power BI.

Five concepts drive everything

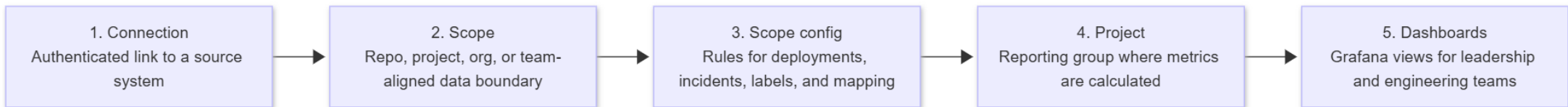
Connection. Authenticated link to a tool (e.g., your GitHub org).

Scope. What to collect within a connection — a repo set, a Jira board.

Scope Config. Rules for how raw data maps to DORA concepts: deployment patterns, incident labels, environment naming.

Project. Logical grouping of scopes — DORA is calculated at this level.

Blueprint. Recurring sync schedule (created automatically).



Cost, ownership, and the fine print

Cost & operations

- Local: free (Docker on a developer laptop)
- Azure: ~\$30–50 USD/month — Azure Container Instances + managed MySQL + Key Vault
- Owned by a DevEx, platform, or engineering productivity team — not individual devs
- Access via PATs or service credentials per data source

Things to know

- Apache Software Foundation *Incubating* — maturing project, not yet top-level
- Some tools need webhooks or custom plugins (e.g., Azure Pipelines)
- Data freshness depends on Blueprint sync cadence (daily default)
- DORA accuracy depends on Scope Config (deployment patterns, incident labels) being set correctly



Part 3

Adopting DevLake

A practical adoption path

Phase 1 — Foundation

~1 week

Local Docker deployment. Connect one or two source tools. Onboard 2 pilot teams. Prove DORA + Copilot data flows end-to-end.

Phase 2 — Expand

weeks 2–4

Move to Azure-hosted instance. Add remaining connectors. Onboard priority teams or business units. Produce first cross-team AI impact views.

Phase 3 — Optimise

month 2

Tune Scope Configs. Build comparison dashboards across teams. Add PR lifecycle metrics. Migrate source hosts if in progress.

Phase 4 — Mature

month 3+

Extend with custom plugins or webhooks for unsupported tools. Integrate incidents. Export to enterprise data warehouse for business blending.

Install & initialise

```
gh extension install DevExpGBB/gh-devlake  
gh devlake init
```

Getting started with gh-devlake

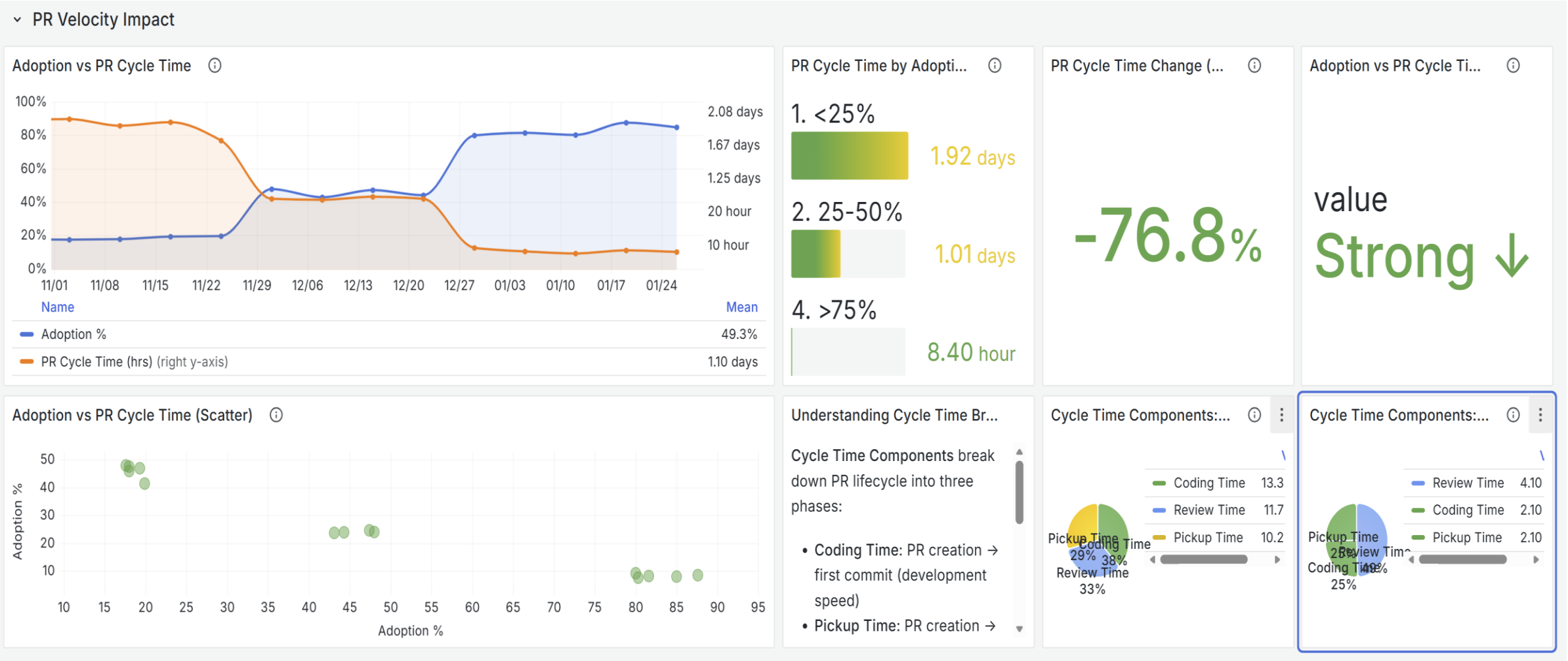
One command — install the GitHub CLI extension

Deploy anywhere — local Docker or Azure via Bicep

Guided wizard — deploy → connect → scope → project → first sync

Dashboards live at <http://localhost:3002> (admin / admin)

Example: PR Velocity Impact



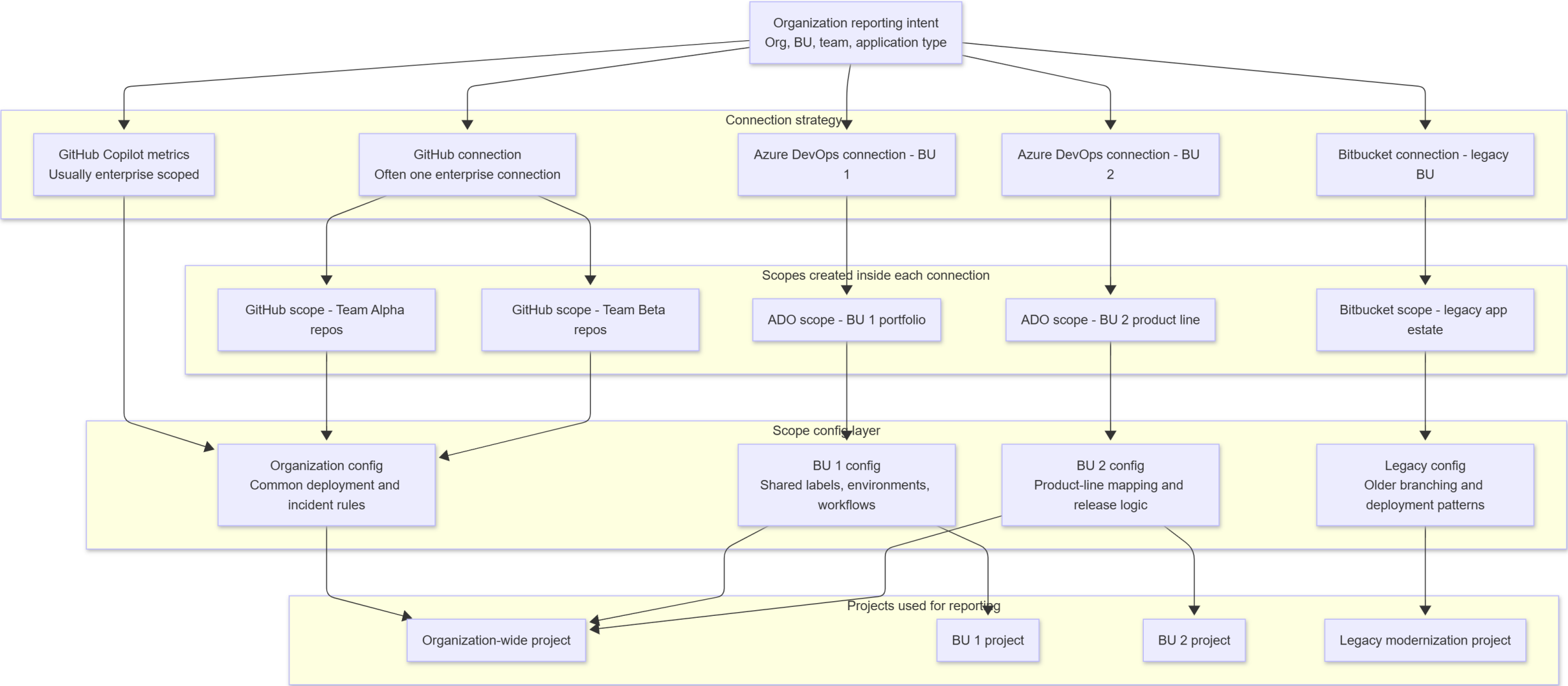
Correlating Copilot adoption with PR cycle time reveals a -76.8% change in cycle time between the lowest and highest adoption tiers. Every panel is a SQL query you can inspect and extend.



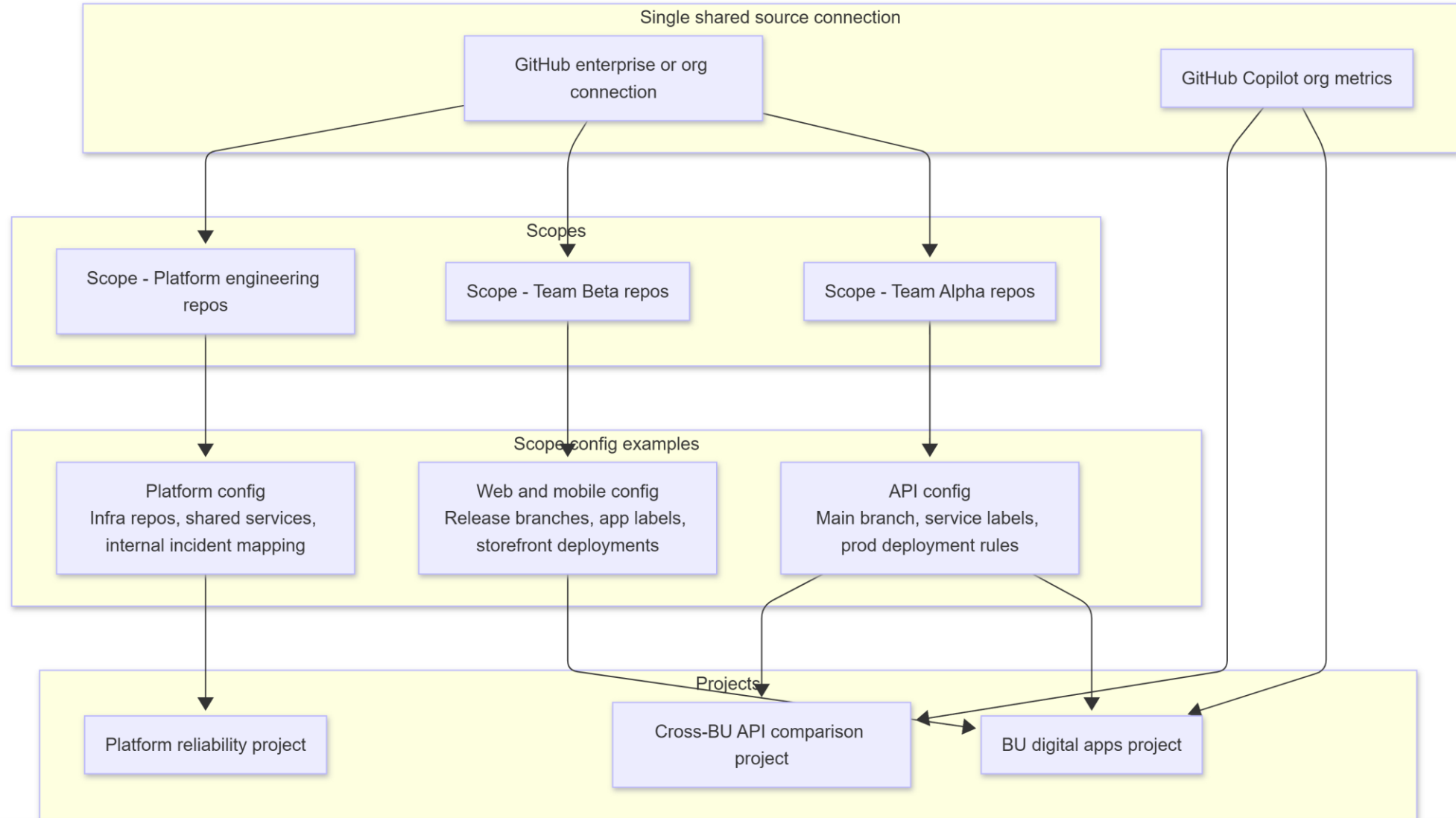
Part 4

Reference Architectures

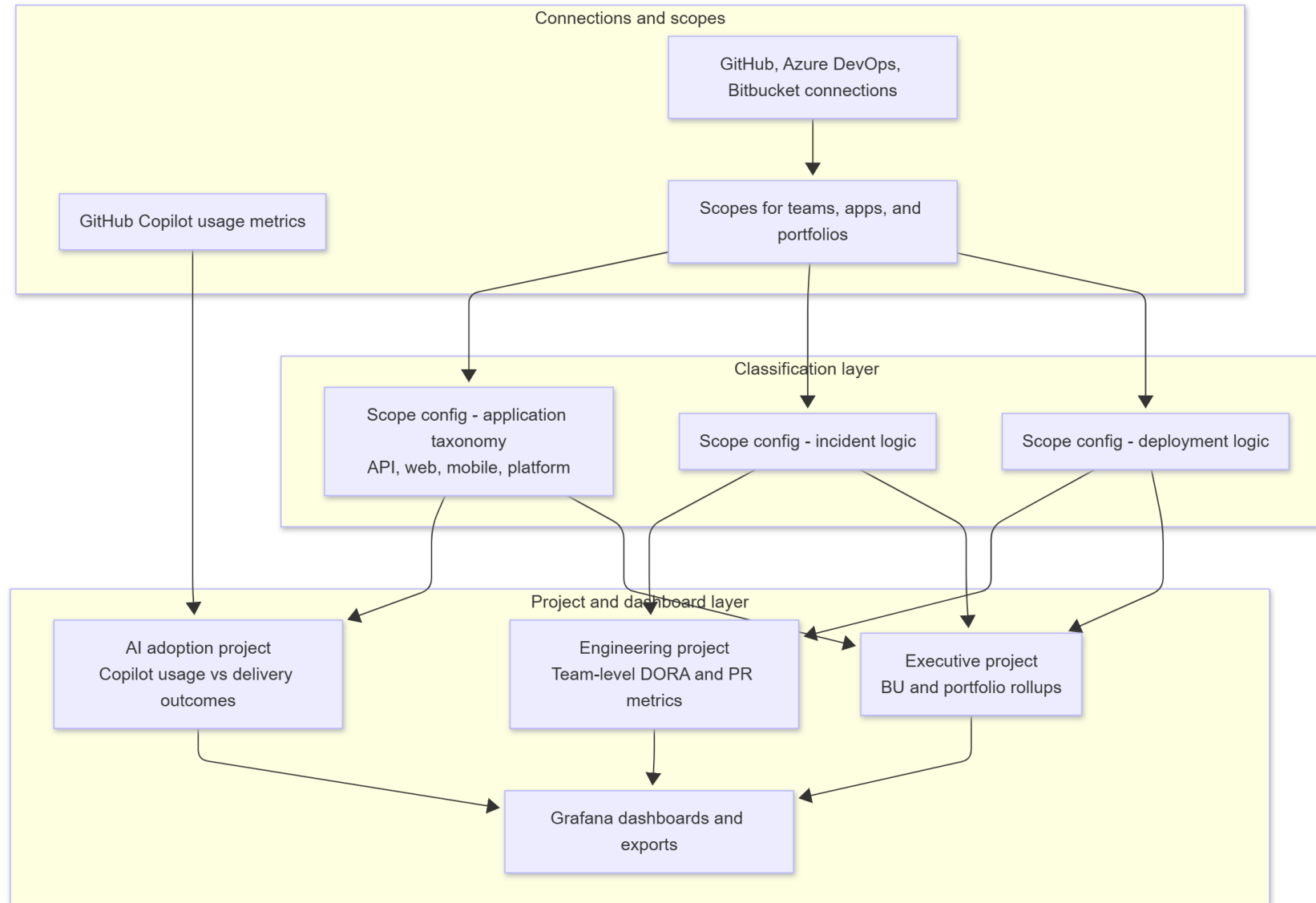
Splicing by Organization and Business Unit



Splicing by Team and Application Type



Splicing Data for Executives & Engineering Reporting



Resources & next steps

Read

- devlake.apache.org
- github.com/apache/incubator-devlake
- DORA & Copilot metric docs

Reference

- DORA research: dora.dev
- Blueprint & Scope Config guide
- Connector catalogue (20+ tools)

Community

- Apache incubator mailing list
- DevLake Slack workspace
- GitHub Discussions & Issues

Deploy

- gh-devlake (GitHub CLI extension)
- Docker Compose (local)
- Azure Bicep templates

Dashboards

- Grafana at localhost:3002
- Default login: admin / admin
- DORA + AI Impact presets included

Next steps

- Run a two-team local pilot
- Agree on DORA + AI metric set
- Identify DevEx/platform owner